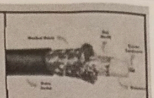
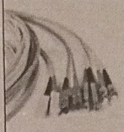
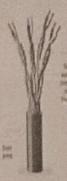
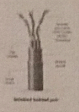
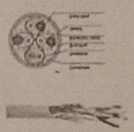
Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile • **Disadvantages** • Limited distance • Cost • Size is bulky	Speed of signal is 500m Television network High speed internet connections	
fibre optics cable	Single mode Multi mode	100Gbps	**Advantages** • High speed • High bandwidth • High security • Long distance **Disadvantages** • Expensive • Requires skilled installers	• Maximum distance of fibre optics cable is around 100meters	

Cable type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/Use	Image
UTP	Category 3	10 bps	**Advantages** • Cheaper in cost • Easy to install as they have a smaller overall diameter. **Disadvantages** • More prone to (EMI) Electromagnetic interference and noise	10Base-T Ethernet	
	Category 5	Up to 100 Mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category6,6a	10Gbps	**Advantages** • Shielded. • Faster than UTP. • Less susceptible to noise and interference **Disadvantages** • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres	
SSTP	Category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100m)	

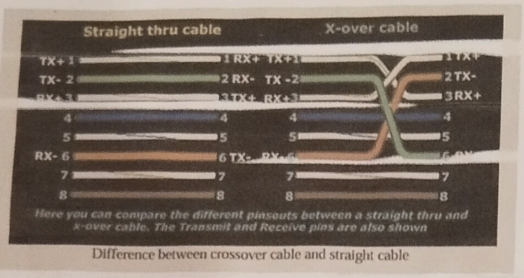
Exp No: 03 Study of different types of
Date: 23/7/26 Network cables.

Aim:-
To understand different types of network cable and make our own ethernet cross-cover cable / straight cable.

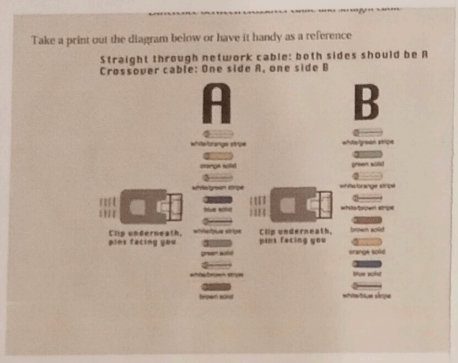
Requirements:-
Different types of cables used in networking are 1) Unshielded Twisted pair (UTP) cable.
2) shielded Twisted pair (STP) cable
3) Coaxial cable
4) Fibre optic cable.

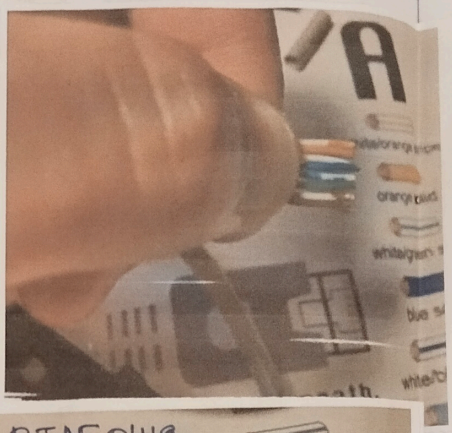
Tools and parts needed to make your own Ethernet cross-over cable / straight cable.
A crimping tool. This is an all in one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
Two RJ45 plugs
optional two plug shields.

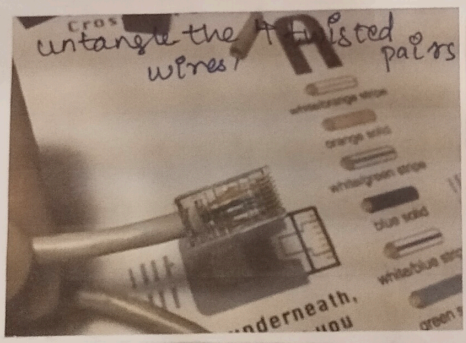
Procedure:-
Step1:- to start construction of the device, begin by shields onto the cable.
Step2:- Next, strip approximately 1.5 cm of cable shielding from both ends the crimping tool has a round area to complete this task.

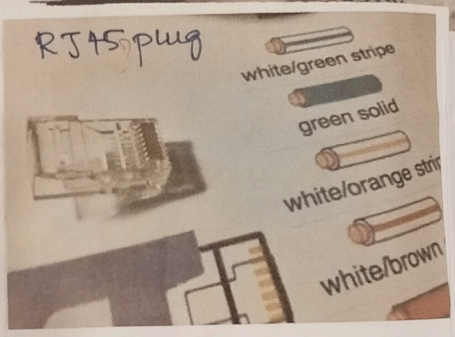


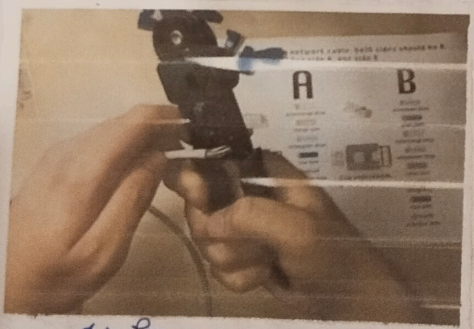
Difference between crossover cable and straight cable





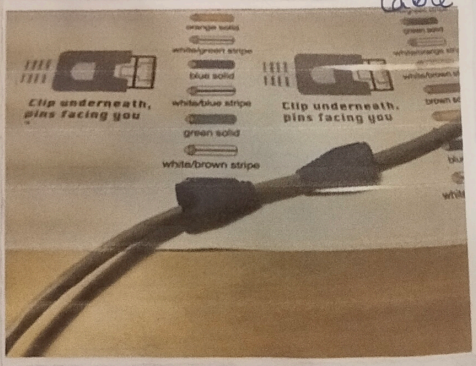






strip 1.5cm of cable shielding from both ends.

threading shields onto the cable



The crimping tool

Step 3:- After, untangle the wires, there shld be four "twisted pairs" Referencing back to the sheet.

Step 4:- Once the order is correct, bunch them together in a line and if there are any that stick out farther than others.

Step 5:- Next, push the cable right in the notch at the end of the plug needs to be just over the cable.

Step 6:- After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7:- Lastly repeat for the other end using diagram B and diagram A.

Result:-

Thus the different types of network cables were studied successfully.